

Lecture 14: Related Rates (Perimeter/Area/Volume Relations)

Goal: If a quantity changes over time, explain how its derivative changes over time.

1 Introduction (Circles)

Problem 1. A circle's area and radius are related by the equation:

$$A = \pi r^2$$

If A and r are changing as time advances, is there any relation between $\frac{dA}{dt}$ and $\frac{dr}{dt}$?



$$A = \pi r^2$$

$$A(t) = \pi [r(t)]^2$$

$$\frac{d}{dt}(A(t)) = \pi \frac{d}{dt}([r(t)]^2)$$

$$\frac{dA}{dt} = \pi \left[2[r(t)] \frac{dr}{dt} \right]$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Problem 2. (a) In the previous example, if $r = 2$ and $\frac{dr}{dt} = 3$, then what is the value of $\frac{dA}{dt}$? Interpret.

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$\left. \frac{dA}{dt} \right|_{r=2} = 2\pi (2)(3) = 12\pi$$

Therefore, when the radius is 2 units long, the area is increasing at a rate of 12π units²/sec.

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(b) In the previous example, if $r = 1$ and $\frac{dA}{dt} = 2\pi$, then what is the value of $\frac{dr}{dt}$?

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$2\pi = 2\pi(1) \frac{dr}{dt}$$

$$\frac{dr}{dt} = 1$$

Therefore, if the radius is 1 unit long, in order for the area to change at a rate of $2\pi \text{ units}^2/\text{sec}$, the radius will need to change at a rate of $1 \frac{\text{unit}}{\text{sec}}$.

Problem 3. The radius of a circle r is increasing at a constant rate 3 cm/min.

(a) Find the rate of change of the area of the circle (A) when the radius is 5 cm.

A = Area of the circle
 r = radius of the circle.



Formula: $A = \pi r^2$ | K: $\frac{dr}{dt} = 3$

NTK: $\frac{dA}{dt}$ when $r = 5$

$$\begin{array}{l|l} A = \pi r^2 \\ \frac{dA}{dt} = 2\pi r \frac{dr}{dt} \end{array} \left| \begin{array}{l} \left[\frac{dA}{dt} \right]_{r=5} r'=3 = 2\pi(5)(3) \\ = 30\pi \end{array} \right| \begin{array}{l} \text{When the radius} \\ \text{is 5 cm, the area} \\ \text{of the circle is} \\ \text{increasing at a} \\ \text{rate of } 30\pi \text{ cm}^2/\text{min.} \end{array}$$

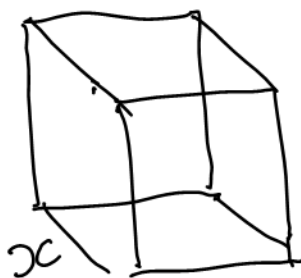
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- (b) Find the rate of change of the circumference of the circle (C) when the radius is 5cm.

2 Rectangular Prisms

Problem 4. The edges of a cube are shrinking at a rate of 10 cm/s.

- (a) How fast is the volume (V) shrinking when each side length is 9cm long?



$V =$ Volume
 $x =$ length of each side

Formula:

$$V = x^3$$

K:

$$\frac{dx}{dt} = -10$$

NTK:

$$\frac{dV}{dt} \text{ when } x=9$$

$$\begin{array}{l|l} V = x^3 & \left. \frac{dV}{dt} \right|_{x=9, x'=-10} = 3(9)^2(-10) \\ \frac{dV}{dt} = 3x^2 \frac{dx}{dt} & = -2430 \\ \text{[Handwritten note: } \frac{d(x^3)}{dx} \cdot \frac{dx}{dt} \text{]} & \end{array} \quad \left| \begin{array}{l} \text{When the side} \\ \text{lengths are 9 cm,} \\ \text{the volume is} \\ \text{decreasing at a} \\ \text{rate of } 2430 \frac{\text{cm}^3}{\text{s}}. \end{array} \right.$$

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- (b) How fast is the surface area (A) shrinking when each side length is 9cm long?



$A =$ Surface Area of the cube

Formula:

$$A = 6x^2$$

K:

$$\frac{dx}{dt} = -10$$

WTK:

$$\frac{dA}{dt} \text{ when } x=9$$

$$A = 6x^2$$

$$\frac{dA}{dt} = 12x \frac{dx}{dt}$$

Here is when you'd plug in numbers.

$$\left. \frac{dA}{dt} \right|_{\substack{x'= -10 \\ x=9}} = 12(9)(-10)$$
$$= -1080$$

When the side lengths are 9 cm, the surface area is decreasing at a rate of $1080 \text{ cm}^2/\text{s}$.

3 Spheres

Problem 5. A balloon is (roughly) a sphere. The balloon deflates and its radius decreases at a rate of 2 cm/s.

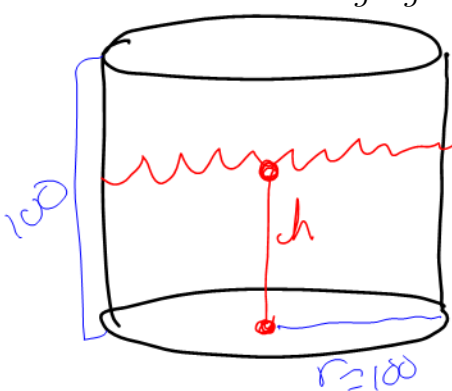
- (a) How fast is the volume (V) shrinking when the radius is 5cm long?

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(b) How fast is the surface area (A) shrinking when the radius is 5cm long?

4 Cylinders

Problem 6. A cylindrical tank with a radius and height of 100 cm stands upright. Water is being drained at a rate of $7 \text{ cm}^3/\text{s}$. How fast is the water level changing when the tank is half empty.



$h = \text{Water Level}$

$V = \text{Volume/Amount of Water}$

Formula:

$$V = \pi r^2 h$$

$$V = \pi (100)^2 h$$

$$V = 10000 \pi h$$

K:

$$\frac{dV}{dt} = -7$$

NTK:

$\frac{dh}{dt}$ when

$$h = \frac{1}{2}(100) = 50$$

$$V = \pi r^2 h$$

$$\frac{dV}{dt} = 10000 \pi \frac{dh}{dt} \quad \left| \frac{dh}{dt} \right|_{V'=-7} = \frac{-7}{10000 \pi} \frac{\text{cm}}{\text{s}}$$
$$\frac{dh}{dt} = \frac{1}{10000 \pi} \frac{dV}{dt} \quad \approx -2.2 \frac{\text{mm}}{\text{s}}$$

5 Cones

Problem 7. Sand pours onto a surface at $15\text{cm}^3/\text{s}$, forming a conical pile with a base diameter that is always equal to the pile's altitude. How fast is the altitude of the pile increasing when the pile is 8cm high?



h = height / altitude

r = radius

V = Volume

Formula:

$$V = \frac{\pi}{3} r^2 h$$

$$V = \frac{\pi}{3} \left(\frac{h}{2}\right)^2 h$$

$$V = \frac{\pi}{12} h^3$$

K:

$$\frac{dV}{dt} = 15$$

$$2r = h$$

NTK:

$$\frac{dh}{dt} \text{ when } h = 8$$

$$\frac{dV}{dt} = \frac{\pi}{12} \left(3h^2 \frac{dh}{dt} \right)$$

$$\frac{dV}{dt} = \frac{\pi}{4} h^2 \frac{dh}{dt}$$

$$15 = \frac{\pi}{4} (8)^2 \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{15}{16\pi} \frac{\text{cm}}{\text{s}}$$